Copenhagen interpretation of quantum mechanics. But people, including the physicists from Copenhagen who purportedly invented this interpretation, disagree on precisely what that label should be taken to describe. We can just refer to it as "standard textbook quantum mechanics."

The idea that these rules represent how reality actually works is, needless to say, outrageous.

What precisely do you mean by a "measurement"? How quickly does it happen? What exactly constitutes a measuring apparatus? Does it need to be human, or have some amount of consciousness, or perhaps the ability to encode information? Or maybe it just has to be macroscopic, and if so how macroscopic does it have to be? When exactly does the measurement occur, and how quickly? How in the world does the wave function collapse so dramatically? If the wave function were very spread out, does the collapse happen faster than the speed of light? And what happens to all the possibilities that were seemingly allowed by the wave function but which we didn't observe? Were they never really there? Do they just vanish into nothingness?

To put things most pointedly: Why do quantum systems evolve smoothly and deterministically according to the Schrödinger equation *as long as we aren't looking at them*, but then dramatically collapse when we do look? How do they know, and why do they care? (Don't worry, we're going to answer all these questions.)

Science, most people think, seeks to understand the natural world. We observe things happening, and science hopes to provide an explanation for what is going on.

In its current textbook formulation, quantum mechanics has failed in this ambition. We don't know what's really going on, or at least the community of professional physicists cannot agree on what it is. What we have instead is a *recipe* that we enshrine in textbooks and teach to our students. Isaac Newton could tell you, starting with the position and velocity of a rock that you have thrown into the air in the Earth's gravitational field, just what the subsequent trajectory of that rock was going to be. Analogously, starting with a quantum system prepared in some